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A simple judgment method of gray property of flames based on spectral analysis and the two-color method for measurements of temperatures and emissivity

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Abstract

A method for judging the gray property of flames based on spectral analysis and the two-color method for determining the temperature and emissivity of a flame has been demonstrated. A calibrated spectrometer system was used for the data acquisition of the radiation intensity profile of a flame over a range of wavelengths (200–1100 nm). The wavelength range that meets the gray body assumption, in which the emissivity can be assumed to be constant, can be determined from the emissivity profile obtained by spectral analysis and the two-color method. Then the temperature and emissivity of the flame and their relative mean square deviations were calculated within that range. Experiments were conducted on solidified gasoline and red phosphorus flames, and pulverized coal-fired flames in a commercial 420 ton/h boiler furnace. The results show that the gasoline flame can be assumed to be a gray body in the range between 550 and 900 nm, for the coal-fired flame the range is between 500 and 1000 nm; while the red phosphorus flame cannot be assumed to be a gray body within the measurement wavelength range. The temperature and emissivity calculation results of coal-fired flames are found to be in reasonable agreement with results using other methods described in the literature. The wavelength interval in the two-color method to calculate temperature is also discussed.
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Keywords: Temperature measurement; Radiation property; Two-color method; Spectroscopic analysis; Flame emissivity

1. Introduction

The measurement of temperature is essential for the development of combustion theory and technology. A number of methods for flame temperature measurement have been studied in the past. Laser spectroscopy methods, such as laser interferometric holography [1], planar laser-induced fluorescence (PLIF) [2], coherent anti-Stokes Raman scattering spectroscopy (CARS) [3] and crossed plane Rayleigh imaging [4], have been reported to measure the flame temperatures for many years. But these methods need expensive equipment and complex optical systems and, most importantly, are not easily used in industrial furnaces.

The traditional spectral method using an infrared radiative pyrometer [5] or a photoelectric pyrometer [6] are also in use today to measure

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temperatures in many environments. But these methods are mostly dependent on the emissivity. To measure the temperature, the emissivity should be set to a fixed value, while, in practice, the actual emissivity value is unknown in many circumstances.

Temperature measurement methods based on image processing techniques have also been widely used [7–12], which usually use the two-color method to calculate temperatures. But some complex optical–electro-mechanical systems are required to separate the monochromatic radiation energies of two wavelengths from the same radiation source [8]. So the two-color method based on a tri-chromatic signal of the image has been adopted [9–12], which can get monochromatic radiation images at different wavelengths through image processing techniques. However, the radiation energy of the tri-chromatic signal is not exactly monochromatic and the wavelengths of the tri-chromatic signal may be different using different image acquisition systems [12]. The two-color method is usually based on the assumption that the emissivity at the two wavelengths is the same, but this assumption may not be examined sometimes, which will affect the veracity of the results.

If the spectral emissivity changes with the wavelength and their relationship is unknown, the two-color method is not suitable for temperature measurement. The multi-wavelength method [13–15] has been developed by many researchers with the assumption that the change of emissivity with wavelength satisfies certain regularity. But currently there is not a uniform emissivity model that can be used in every situation. Some researchers have attempted to use intelligent methods to figure out the problem [16], but these are not fully developed.

If the monochromatic emissivity of a body is independent of wavelength, the body can be assumed to be a gray source. Though for a flame dominated by soot, a gray assumption would not be appropriate because the absorption coefficient varies as the reciprocal of wavelength due to the near-Rayleigh size of the particles [17–19]. In practice, many researchers [8–12] assumed the flame to be a gray source to simplify the calculation when the wavelength interval is close enough, and this assumption was mainly used for flames in large-scale furnace conditions, which are different from optically thin small laboratory soot dominated flames [20]. For real objects, the gray body assumption may not be tenable or may only be appropriate over a limited wavelength range. Therefore, it is desirable to find a method to judge whether an object can be assumed to be a grey body and over what wavelength range the gray property is applicable. The spectrometer measurement is an appropriate method by which a large range of spectral intensity information can be obtained.

The fiber-optic spectrometer system introduced here has been widely used to measure flame temperature by many researchers because of its advantages including portability, small dimensions, and geometric versatility. For example, Keyvan et al. [21,22] determined the temperature from high temperature natural gas-fired furnaces based on spectral intensity analysis in the visible spectral range and near-IR range. Cai et al. [23] used least square fitting of the coal flames emission spectra to a theoretical gray body emission with temperature and emissivity as the fitting parameters and compared the fit curve to the measured curve to judge the gray assumption based on the fitting quality. This method would be inappropriate for a non-gray radiation source.

In this paper, a simple method is developed to judge the assumption of gray radiation of different flames based on spectral radiation information obtained by a fiber-optic spectrometer system in the visible spectrum. Then we can get the temperature and emissivity of flames using the two-color method once the gray property is confirmed. The spectral profiles of solidified gasoline and red phosphorus flames are analyzed; experiments are also conducted on a 420 ton/h tangentially coal-fired boiler to analyze the coal-fired flames. The measurement principle and the experimental results are described in the following sections.

2. Measurement principle

The radiation intensity for a given wavelength can be calculated using Wien's Law of Radiation (including the emissivity term) for a temperature range from 800 to 2000 K and wavelength range from 300 to 1000 nm [24]:

$$I(\lambda, T) = \frac{1}{\pi}\varepsilon(\lambda)\frac{C_1}{\lambda^5}\exp\left(-\frac{C_2}{\lambda T}\right) = \varepsilon(\lambda)I_b(\lambda, T) \quad (1)$$

where $I(\lambda, T)$ is the monochromatic radiation intensity at a given wavelength λ and a given temperature T, $I_b(\lambda, T)$ is the monochromatic blackbody radiation intensity, $\varepsilon(\lambda)$ is the monochromatic emissivity for a given wavelength λ, T is the absolute temperature, $C_1 = 2\pi hc^2$, $C_2 = hc/k$ and the constants h, k and c are Plank's constant, the Boltzmann constant and the speed of light, respectively.

If two monochromatic radiation intensities $I(\lambda, T)$, $I(\lambda + \Delta\lambda, T)$ emitted from one point at two different wavelengths λ and $\lambda + \Delta\lambda$ can be obtained, dividing one equation by the other, the following two-color equation is obtained:

$$\frac{I(\lambda, T)}{I(\lambda + \Delta\lambda, T)} = \frac{k_\lambda S(\lambda, T)}{k_{\lambda+\Delta\lambda} S(\lambda + \Delta\lambda, T)} = \left(\frac{\varepsilon_\lambda}{\varepsilon_{\lambda+\Delta\lambda}}\right)\frac{(\lambda + \Delta\lambda)^5}{\lambda^5} \times \exp\left[-\frac{C_2}{T}\left(\frac{1}{\lambda} - \frac{1}{\lambda + \Delta\lambda}\right)\right] \quad (2)$$

In this formula, for wavelength λ, the measured intensity is $S(\lambda, T)$ and the emissivity is ε_λ, and for wavelength $\lambda + \Delta\lambda$, $S(\lambda + \Delta\lambda, T)$ is the measured intensity and $\varepsilon_{\lambda+\Delta\lambda}$ is the emissivity. The calibration factors $k_\lambda, k_{\lambda+\Delta\lambda}$ in Eq. (2) relate to grating efficiency and losses in the fiber-optic cable and are derived from calibration using the blackbody furnace detailed below. If $\Delta\lambda$ is sufficiently small, the ratio $\varepsilon_\lambda/\varepsilon_{\lambda+\Delta\lambda}$ could be assumed to be 1, which simplifies the equation further. But $\Delta\lambda$ cannot be too small, or it could cause a large calculation error. The proper difference $\Delta\lambda$ between the two wavelengths will be discussed below.

Based on the assumption of a constant emissivity, the projection temperature T at wavelengths λ and $\lambda + \Delta\lambda$ is given by

$$T = -C_2\left(\frac{1}{\lambda} - \frac{1}{\lambda + \Delta\lambda}\right) \Big/ \ln\left[\frac{I(\lambda, T)}{I(\lambda + \Delta\lambda, T)}\frac{\lambda^5}{(\lambda + \Delta\lambda)^5}\right] \tag{3}$$

After the temperature T is obtained, the monochromatic emissivity $\varepsilon(\lambda)$ can be found from the following simple equation, the ratio of the radiation intensity of an object to the blackbody intensity at the same temperature

$$\varepsilon(\lambda) = I(\lambda, T)/I_b(\lambda, T) \tag{4}$$

For the spectrometer system, the radiation intensities of many pairs of wavelengths λ and $\lambda + \Delta\lambda$ can be found simultaneously, then the temperature and emissivity profiles based on different pairs of radiation intensity and wavelength are calculated using Eqs. (3) and (4). The average temperature T_a, the average emissivity ε_a, the relative mean square deviation of temperature σ_T and the relative mean square deviation of emissivity σ_ε are introduced to characterize the fluctuation of the temperature and emissivity, where σ_T, σ_ε are respectively the ratio of mean square deviation of temperature and emissivity to T_a, ε_a.

Once the wavelength range meeting the gray property is determined based on the emissivity profile, more accurate temperatures could be calculated by the two-color method within this wavelength range.

The procedure of the measurement is summarized below:

(i) Capture spectral intensity profile $S(\lambda, T)$ using the spectrometer system.
(ii) Get the calibrated radiation intensity profile $I(\lambda, T)$ according to the calibration factor discussed below.
(iii) Obtain the temperature and emissivity profile from the intensity profile $I(\lambda, T)$ by using Eqs. (3) and (4) and their relative mean square deviations.
(iv) Determine the wavelength range meeting the gray property from the emissivity profile.
(v) Calculate the average temperature T_a and emissivity ε_a within the validated gray emission range and compare the theoretical intensity with T_a and ε_a as parameters to the measured intensity.

3. The spectrometer system and its calibration

The spectrometer system consists of a portable computer, a spectrometer and a fiber-optic cable (with a collimating lens). An AvaSpec-2048 Fiber-Optic Spectrometer with 2048 pixel CCD Detector Array is used to process the incoming light data. A choice of 15 different gratings with different dispersion and blaze angles enable applications in the 200–1100 nm range. Connection of the spectrometer to the portable computer is accomplished with a USB2 interface. The collimating lens, COL-UV/VIS, screws onto the end of the fiber optic entrance connector and converts the divergent beam of radiation into a parallel beam. This combination allows integration times from 1.1 ms to 600 s. The integration time was 10 ms in all measured spectral data presented in this paper.

The spectrometer system could get the monochromatic radiation energies within a certain wavelength range, but the output of the spectrometer system is just a voltage value $S(\lambda, T)$ converted from the radiation signal through photoelectric conversion. So it is necessary to calibrate the output of the spectrometer system to get the monochromatic radiation intensity $I(\lambda, T)$. A blackbody furnace with a temperature range from 1000 to 2500 K (with temperature errors within ± 5 K) was used to calibrate the spectrometer system to get the calibration coefficients. These calibration coefficients are determined by dividing the actual measured intensity $S(\lambda, T)$ of the spectrometer system by the monochromatic blackbody radiation intensity $I_b(\lambda, T)$ when the temperature of the blackbody furnace is T, i.e.

$$k_\lambda = I_b(\lambda, T)/S(\lambda, T) \tag{5}$$

Plotting normalized k_λ as a function of λ yields the characteristic calibration profiles within a wavelength range from 500 to 1100 nm at the five temperatures of 1560, 1580, 1600, 1620, and 1640 K, as shown in Fig. 1. The characteristic calibration profiles at the different temperatures are consistent with each other, so the characteristic calibration factors are independent of temperature.

4. Experimental results and analysis

4.1. Wavelength interval in two-color method

The wavelength interval $\Delta\lambda$ is an important factor in the two-color method, which has been paid great attention [25]. The intensity profile

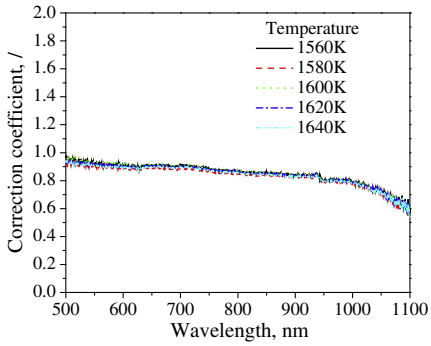

Fig. 1. The characteristic calibration curves for the spectrometer at five temperatures.

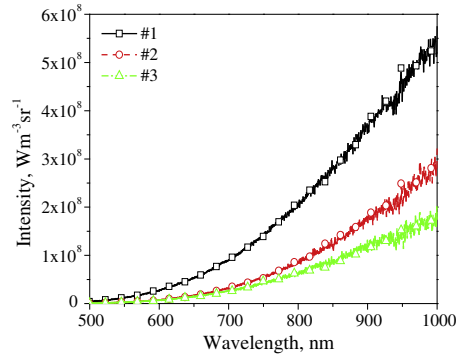

Fig. 2. Three calibrated intensity profiles of a gasoline flame between 500 and 1000 nm.

for a blackbody furnace at 1503 K is taken as a simple profile to find out the proper wavelength interval, 20 different wavelength intervals, ranging from 5 to 100 nm, are used to get the temperature profile. When the wavelength interval is 10 nm, the temperature profile fluctuates widely with the wavelength and σ_T reaches 3.8%. The temperature profile has a small fluctuation with wavelength when the wavelength interval is larger than 30 nm. The relative mean square deviation also decreases as the wavelength interval increases, being less than 1% when the interval is 40 and 60 nm. The average temperature is the same when the interval is 40 and 60 nm, which means there is no difference in the result when the interval is greater than 40 nm, which is also demonstrated by the following experimental results. Therefore, to ensure the accuracy of the calculation, considering the principle of the two-color method, the wavelength interval should be at least greater than 30 nm. In this paper, the wavelength interval has been 40 nm in all calculations.

4.2. Solidified gasoline flame experiment

Experiments were conducted at first on a solidified gasoline flame whose physical properties are almost the same as those of liquid gasoline, and a 20-mm diameter cylindrical block of solidified gasoline with length of 30-mm was placed and fired on a simple open metal platform. The front end of the fiber-optic cable was about 20 mm away from the flame to protect the cable. The temperature and emissivity profiles were recorded and the wavelength range that meets the gray body condition is to be determined.

Three radiation intensity profiles of a solidified gasoline flame were produced based on calibrated measured intensity profiles between 500 and 1000 nm, as shown in Fig. 2. The wavelength range for calculating temperatures was chosen based on the following principles. Firstly, it is necessary to avoid any chemiluminescence emission/absorption lines emitted in the flame. For example, a fluctuation near 950 nm in the intensity profile in Fig. 2 should be avoided in temperature calculations. Secondly, high SNR should be ensured for the measurement data, which means the beginning and the end parts of the measured spectral range should not be selected. For example, the intensity over the wavelength ranges below 500 nm and above 1000 nm is either weak or has low SNR, so these regions should not be included in temperature calculations.

To help minimize the calculation error, 20 spectra are averaged to minimize some variations in the intensities. The temperature profile of intensity #1 in Fig. 3 is calculated using multiple pairs of wavelengths between 500 and 900 nm with an interval of 40 nm as seen in Fig. 3. As shown in Fig. 3, the temperature profiles between 550 and 800 nm are slightly fluctuating but can be regarded as generally flat; the large fluctuation exists near 900 nm because of the fluctuation of intensity profile near 950 nm. The σ_T is only 1.07%, so the average temperature 1732 K could be set as a representative temperature to calculate the emissivity profile also shown in Fig. 3. The emissivity profile is nearly constant between 550 and 900 nm and has only a little fluctuation near 950 nm, while σ_ε is only 0.85%. Therefore, the emissivity in this wavelength range can be

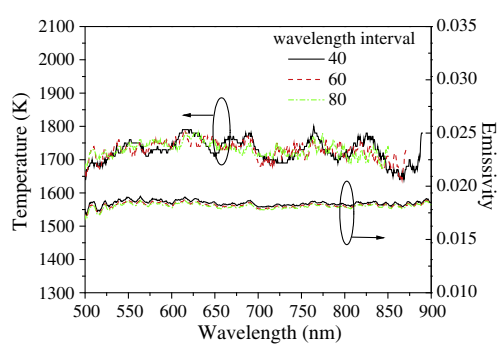

Fig. 3. Temperature and emissivity profiles for the gasoline flame at three wavelength intervals.

regarded as independent of wavelength. In other words, the solidified gasoline flame meets the gray body condition between 550 and 900 nm. The temperature and emissivity profiles at the other two wavelength intervals 60 and 80 nm are also shown in Fig. 3. It shows that the temperature and emissivity results are independent of the wavelength interval. The temperature and emissivity profiles for the other two intensity profiles in Fig. 2 can also be obtained, and their average temperatures between 550 and 800 nm are 1550 K and 1643 K, the σ_T and σ_ε are respectively 1.17% and 1.31%.

4.3. Coal-fired furnace flame experiment

The industrial experimental flame measurements were made at a 420 ton/h tangentially-fired coal-burning boiler. The boiler furnace with horizontal dimensions of 9980 mm (width) × 9980 mm (depth) is shown schematically in Fig. 4. The boiler has two layers of burners, and each layer of four burners is arranged in the four corners of the furnace. Eight measurement points are available in the corners of the furnace, as can also be seen in Fig. 4.

The calibrated intensity profiles between 500 and 1000 nm at four of the eight measurement points of the boiler are shown in Fig. 5. The intensity profiles of #1, #2, #4 and #6 measurement points are taken as examples to analyze the spectral information and calculate the temperature and emissivity. The wavelength range between 500 and 900 nm is selected as the proper range for temperature calculations, as seen in Fig. 6.

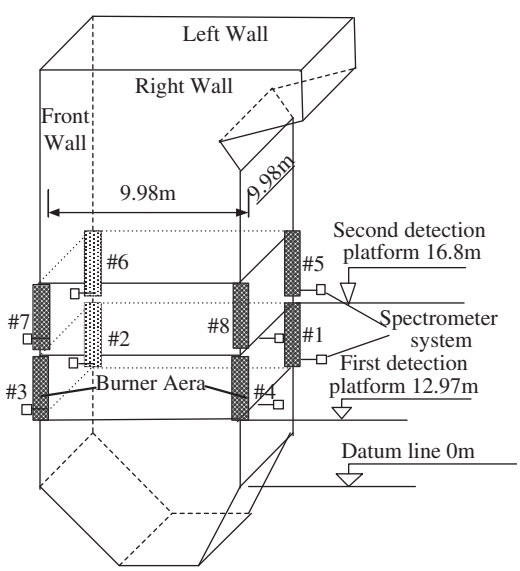

Fig. 4. Schematic diagram of the furnace and locations of the eight measurement points.

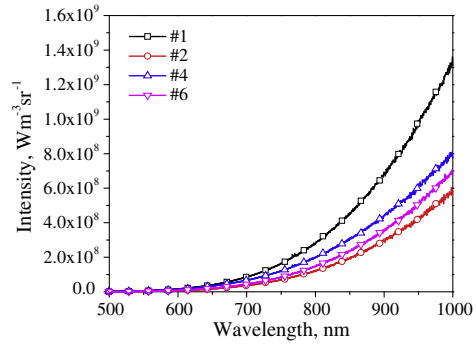

Fig. 5. Calibrated intensity profiles of coal-fired flames measured at four of the eight measurement points.

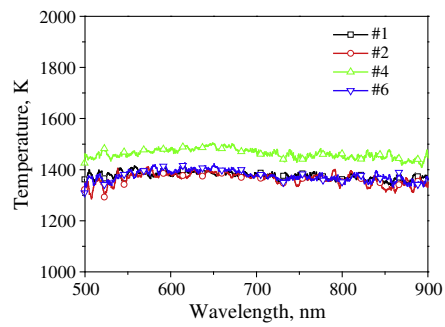

Fig. 6. Temperature profiles of coal-fired flames at four measurement points.

The characteristic temperatures of #1, #2, #4 and #6 obtained by the average temperatures over wavelengths between 500 and 900 nm are 1372, 1362, 1454 and 1367 K, and σ_T's are only 2.33%, 1.62%, 2.21% and 2.27%. Using the characteristic temperature, the emissivity profiles between 500 and 1000 nm can be obtained easily, and these are shown in Fig. 7. In Fig. 7, σ_ε's of the emissivity profiles between 500 and 1000 nm are only 2.30%, 2.49%, 1.97% and 2.23%, which means the coal

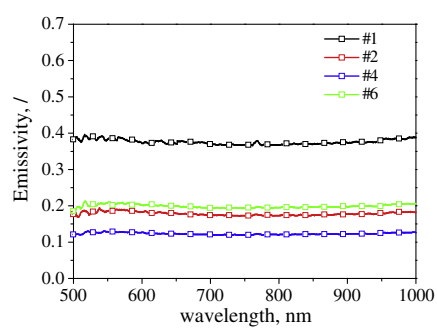

Fig. 7. Emissivity profiles of coal-fired flames at four measurement points.

Table 1
Temperatures and emissivity, and their relative mean square deviations (RMS) between 500 and 1000 nm calculated for every measurement point of the coal-fired boiler furnace and a comparison of temperatures with those obtained by the method used by Cai et al. [23].

Items	Measurement points							
	#1	#2	#3	#4	#5	#6	#7	#8
T_a (K)	1372	1362	1323	1454	1443	1367	1418	1440
σ_T (%)	2.33	1.62	2.85	2.21	4.32	2.27	8.21	3.27
ε_a	0.389	0.188	0.204	0.132	0.036	0.214	0.017	0.219
σ_ε (%)	2.3	2.49	3.05	1.97	3.15	2.23	5.88	3.19
Temperature [Cai]	1371	1361	1327	1454	1450	1367	1440	1439
RMS (%)	0.08	0.06	0.30	0.02	0.48	0.02	1.51	0.10
Emissivity [Cai]	0.390	0.184	0.198	0.129	0.035	0.211	0.018	0.217
RMS (%)	0.25	2.12	2.94	2.27	2.77	1.40	5.88	0.91

flame can be assumed to be a gray body in this wavelength range, which is consistent with the conclusion of Cai et al. [23].

The other intensity profiles can also be analyzed by the same method to calculate the final temperature and emissivity; and the relative mean square deviations of temperature and emissivity between 500 and 1000 nm are shown in Table 1. The σ_T and σ_ε are smaller than 5% and 4% respectively except for the result of #7. The reason is that the SNR of intensity profile of #7 is low; the calculation error is therefore correspondingly high. A comparison between the final temperature and emissivity calculated by the present method and the characteristic temperature and emissivity obtained using the method of Cai et al. [23] is also shown in Table 1. It can be seen that the relative deviations of temperature and emissivity between these two methods are smaller than 0.5% and 5% respectively, except again for the result of #7.

4.4. Red phosphorus flame experiment

The experimental conditions, and the shape and size of the red phosphorus were the same as for the solidified gasoline. The method of dealing with the intensity profiles for the red phosphorus flame is the same as discussed above. Figure 8 shows three calibrated intensity profiles for the red phosphorus flame between 500 and 1000 nm. Figure 9 shows the temperature profiles between 500 and 900 nm calculated using the two-color method. The average temperature between 500 and 600 nm is set as the characteristic temperature to calculate the emissivity profile, shown in Fig. 10. As shown in Fig. 10, the emissivity profiles change with the wavelength, which means that the red phosphorus flame cannot be assumed to act as a gray body between 500 and 1000 nm. Though the final temperature cannot be obtained, the distribution of monochromatic emissivity can

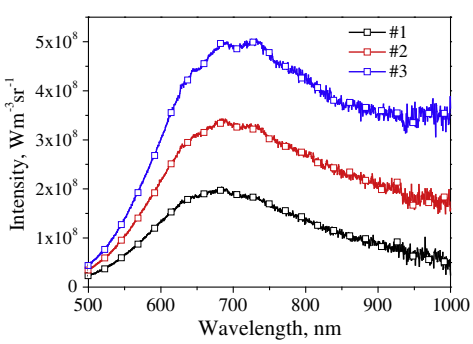

Fig. 8. Intensity profiles of a red phosphorus flame between 500 and 1000 nm.

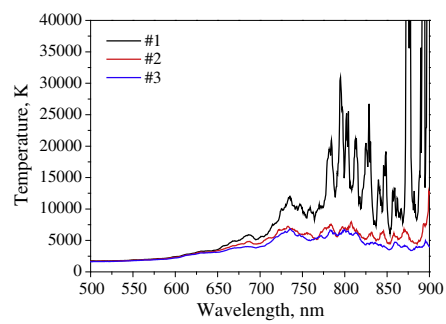

Fig. 9. Temperature profiles of a red phosphorus flame.

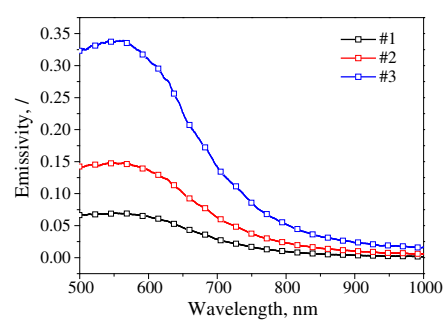

Fig. 10. Emissivity profiles of a red phosphorus flame.

be used as a reference for other temperature measurement methods, such as the multi-wavelength method. The results demonstrate that the temperature of the red phosphorus flame cannot be obtained using the method based on the gray body assumption. For example if, for the calibrated intensity profile #2 in Fig. 8, Cai's method [23] is used to calculate characteristic temperature within the wavelength range (500–1000 nm), the result is 4604 K, which is clearly wrong. In this case, the measurement of flame temperature by the spectral analysis needs further study. For example, it is necessary to take the flame as an inhomogeneous medium with inhomogeneous, non-gray properties.

5. Conclusions

A simple method to find the wavelength range that meets the gray body condition of a spectral intensity profile; and to then get the temperature and emissivity of the spectra was presented in this paper. Experiments for solidified gasoline and red phosphorus flames, and a commercial coal-burning furnace are reported here. The gasoline flame, for the wavelength range between 550 and 900 nm, can be assumed to be a gray body. The coal-burning flame meets the gray condition for the wavelength range between 500 and 1000 nm, which is similar to the result obtained by Cai et al. [23]. The comparison between the results obtained using this method and those obtained by the method used by Cai et al. demonstrates the validity of the measurement results. The red phosphorus flame cannot be assumed to be a gray body between 200 and 1100 nm in the spectrometer measuring range studied in this paper.

The wavelength interval in the two-color method is an important parameter in calculating temperature. This paper discusses the parameter and concludes that the wavelength interval should be at least 30 nm to ensure the accuracy of the calculated temperature. Selecting the appropriate wavelength region to calculate the final temperature and emissivity is also a typical challenge. The relative mean square deviation is used to analyze the smoothness of the distribution curves of temperature and emissivity. The final temperature and emissivity calculated in the wavelength region meeting the gray condition is independent of wavelength interval as long as the interval is larger than 30 nm; the result will then be more accurate.

Future work will involve analyzing the temperature and emissivity of spectral intensity profiles that do not meet the gray body condition, such as those for the red phosphorus flame.

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